

Problem 2 a)

(a) $y \in \mathbb{R}^p$

$\Sigma = y^T y$

$y_{p \times 1}$

$\Sigma = E[(y - \bar{y})(y - \bar{y})^T]$

~~where $\bar{y} = \frac{1}{p} \sum_{i=1}^p y_i$~~

where $\bar{y} = E[y]$

A matrix Σ is called positive semi-definite if

$$v^T \Sigma v \geq 0 \quad \forall \quad v_{p \times 1}$$

$$v^T \Sigma v = v^T E[(y - \bar{y})(y - \bar{y})^T] v$$

$$= E[v^T (y - \bar{y})(y - \bar{y})^T v]$$

$$(y - \bar{y})^T = 1 \times p \quad v = p \times 1$$

$$\therefore (y - \bar{y})^T v = 1 \times 1$$

Similarly $v^T (y - \bar{y}) = 1 \times 1$

$$\therefore v^T \Sigma v = E[(v^T (y - \bar{y}))^2] \geq 0$$

Let v be the eigen vector of Σ

$$\therefore \Sigma v = \lambda v$$

$$v^T \Sigma v = \lambda v^T v$$

$$= \lambda \|v\|^2$$

$$\therefore \lambda \|v\|^2 \geq 0$$

$$\Rightarrow \underline{\underline{\lambda \geq 0}}$$

(eigen values are non-negative)

b) $A = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} \quad \rho \in (0, 1)$

Let v be an eigen vector
 $Av = \lambda v$

$$\Rightarrow (A - \lambda I)v = 0$$

$$\therefore \det(A - \lambda I) = 0 \quad (\text{characteristic polynomial})$$

$$\Rightarrow (1 - \lambda)^2 - \rho^2 = 0$$

$$\Rightarrow 1 - \lambda = \pm \rho$$

$$\Rightarrow \lambda = 1 - \rho \quad \text{or} \quad 1 + \rho$$

$$A \begin{pmatrix} v_{11} \\ v_{12} \end{pmatrix} = (1 - \rho) \begin{pmatrix} v_{11} \\ v_{12} \end{pmatrix}$$

$$v_{11} + \rho v_{12} = (1 - \rho)v_{11} \Rightarrow v_{12} = -v_{11}$$

$$\rho v_{11} + v_{12} = (1 - \rho)v_{12}$$

$$\|v_1\| = 1$$

$$\Rightarrow v_{11}^2 + v_{12}^2 = 1 \Rightarrow v_{11} = \frac{1}{\sqrt{2}} \quad ; \quad v_{12} = -\frac{1}{\sqrt{2}}$$

$$A \begin{pmatrix} v_{21} \\ v_{22} \end{pmatrix} = (1 + \rho) \begin{pmatrix} v_{21} \\ v_{22} \end{pmatrix} \Rightarrow v_{21} = v_{22}$$

$$\|v_2\| = 1$$

$$\Rightarrow v_{21}^2 + v_{22}^2 = 1 \Rightarrow v_{21} = \frac{1}{\sqrt{2}} \quad ; \quad v_{22} = \frac{1}{\sqrt{2}}$$

$$\lambda_1 = 1 - \rho \rightarrow v_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$$

$$\lambda_2 = 1 + \rho \rightarrow v_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

(c) $AV_1 = \lambda_1 V_1$, let A is $n \times n$

$$AV_2 = \lambda_2 V_2$$

$$\vdots$$

$$AV_n = \lambda_n V_n$$

let $V = (V_1 \ V_2 \ \dots \ V_n)$

$$AV = A(V_1 \ V_2 \ \dots \ V_n)$$

$$= (AV_1 \ AV_2 \ \dots \ AV_n)$$

$$= (\lambda_1 V_1 \ \lambda_2 V_2 \ \dots \ \lambda_n V_n)$$

$$= V \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \lambda_n \end{bmatrix} = V \Lambda$$

$$AV = VA$$

$$\cancel{AVV^{-1} = VAV^{-1}}$$

$$\Rightarrow AVV^{-1} = VAV^{-1}$$

$$\Rightarrow A = VAV^{-1}$$

But V is orthogonal matrix (orthonormal)

$$\text{Hence } V^{-1} = V^T \quad (VV^T = I)$$

$$\therefore A = V\Lambda V^T$$

$$(d) \quad A^{-1} = (V\Lambda V^T)^{-1}$$

$$= (V^T)^{-1} \Lambda^{-1} V^{-1}$$

$$= V\Lambda^{-1}V^T$$

$\Rightarrow A$ & A^{-1} have same eigen vectors

the eigen values of A^{-1} is $1/\text{eigenvalue of } A$

For A to be invertible, all eigen values must be non-zero.